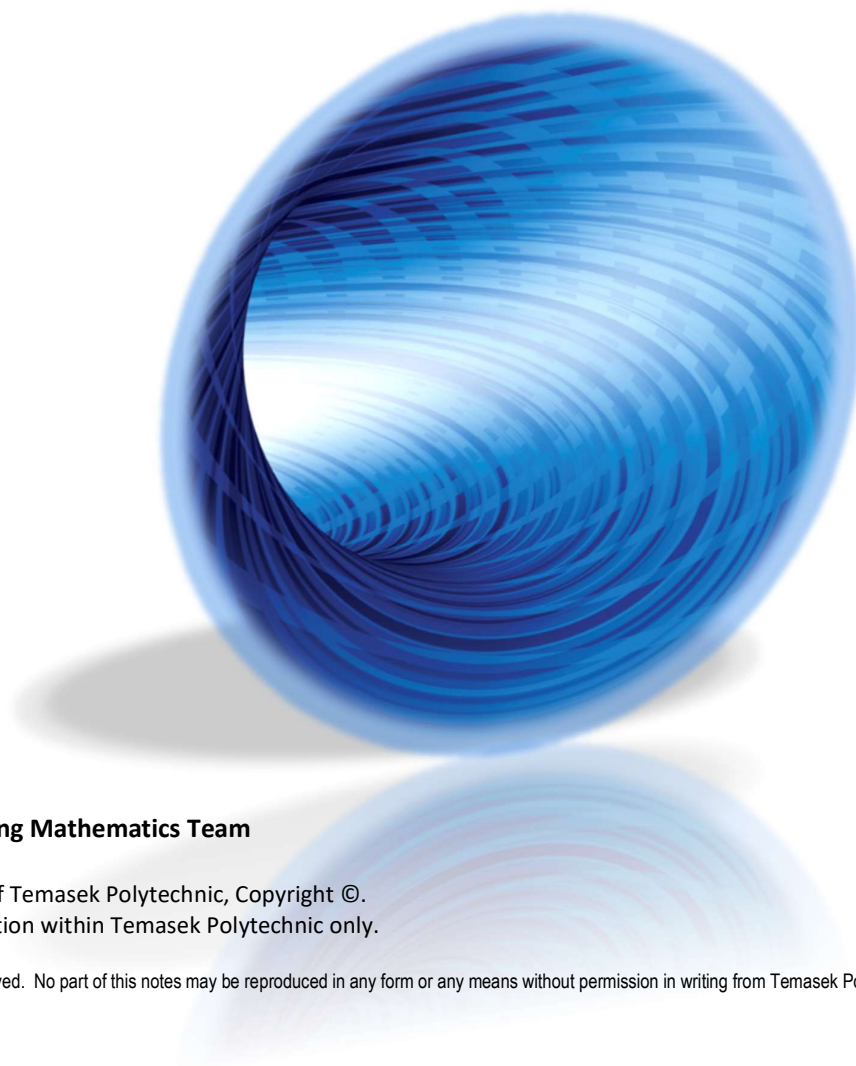


University Preparatory Program (Mathematics) Vectors



Engineering Mathematics Team

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4. Vectors

Learning Objectives

Upon completion of this lesson, you should be able to:

1. Calculate the norm of a vector
2. Calculate the dot product and cross product of 3D vectors
3. Construct equations for lines and planes in $\mathbb{R}^3$
4. Calculate distances from a point to a line, and from a point to a plane
5. Identify various kinds of quadric surfaces

4.1 2D and 3D Vectors

A **vector** in 2D plane or 3D space is a directed line segment, which has both magnitude and direction. A **scalar**, on the other hand, only has magnitude. Throughout this chapter, scalars refer to real numbers. Figure 4.1.1 shows the geometric representation of the vector $\mathbf{v}$ as an arrow with a fixed magnitude and direction. The vector $\mathbf{v}$ can also be denoted by $\vec{v}$, or the directed line segment $\overrightarrow{PQ}$.

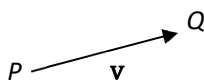


Figure 4.1.1. A vector $\mathbf{v}$ is a directed line segment $\overrightarrow{PQ}$

A two-dimensional (2D) vector $\mathbf{p}$ can be sketched on the 2D rectangular coordinate system by drawing a line segment from the origin $(0, 0)$ to a coordinate point $P(a, b)$ as shown in Figure 4.1.2.

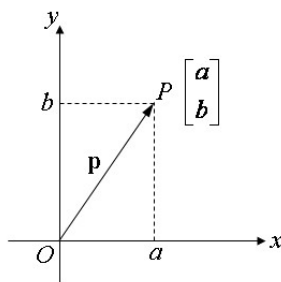


Figure 4.1.2. A vector in 2D rectangular coordinate system

The **norm** (or magnitude) of a vector $\mathbf{p}$ is the length of the directed segment, denoted by $\|\mathbf{p}\|$ or $\|\vec{p}\|$ or $\|\overrightarrow{OP}\|$. The direction of a vector is the angle between the line segment and a reference line. Referring to Figure 4.1.2, the norm and direction of the vector $\mathbf{p}$ is given as

$$(2D) \text{ Norm: } \|\mathbf{p}\| = \sqrt{a^2 + b^2} \quad (4.1)$$

$$(2D) \text{ Direction: } \tan \theta = \frac{b}{a} \text{ for } a \neq 0, \quad (4.2)$$

where θ is the angle between the line segment and positive x -axis

Algebraically, vectors are represented as column matrices. Referring to Figure 4.1.2, we represent the vector $\mathbf{p}$ as the column matrix $\begin{bmatrix} a \\ b \end{bmatrix}$. Hence, it is possible to define the **zero (null) vector**, $\mathbf{0}$, as $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. The null vector has zero norm and no direction.

Given two vectors $\mathbf{p} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} c \\ d \end{bmatrix}$, the addition $\mathbf{p} + \mathbf{q}$ and the scaling $k\mathbf{p}$ for any scalar k are done in the same way as matrix algebra:

$$\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + c \\ b + d \end{bmatrix} \text{ and } k \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ kb \end{bmatrix}.$$

For example, $\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ shown in Figure 4.1.3(a), and $2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ shown in Figure 4.1.3(b).

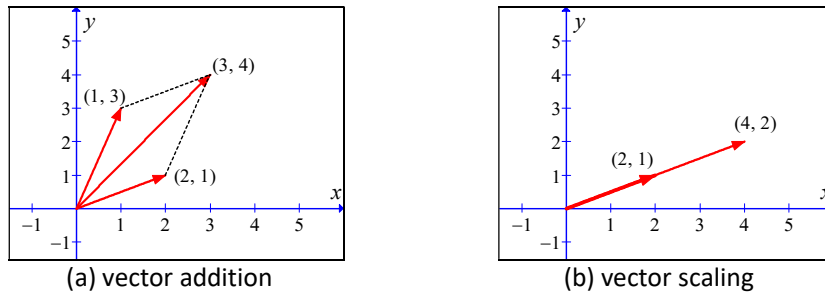


Figure 4.1.3. Vector addition and scaling on the 2D rectangular coordinate system

The scaling of a vector $\mathbf{v}$ with a nonzero scalar k will yield a new vector $k\mathbf{v}$ that has either the same direction (for $k > 0$) or opposite direction (for $k < 0$). Any vector with norm 1 is called a **unit vector**. The unit vector in the direction of $\mathbf{v}$ is denoted by $\hat{\mathbf{v}}$ (read: “vee hat”), and is given as

Unit Vector: $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$ (4.3)

The vector addition has similar properties as the number addition:

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutative)
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{v} + (\mathbf{u} + \mathbf{w})$ (associative)

The vector subtraction $\mathbf{u} - \mathbf{v}$ is the vector addition of $\mathbf{u}$ and $-\mathbf{v}$.

A three-dimensional (3D) vector $\mathbf{p}$ is an extension of 2D vector and can be sketched on the 3D rectangular coordinate system by drawing a line segment from the origin $(0, 0, 0)$ to a point $P(a, b, c)$ as shown in Figure 4.1.4.

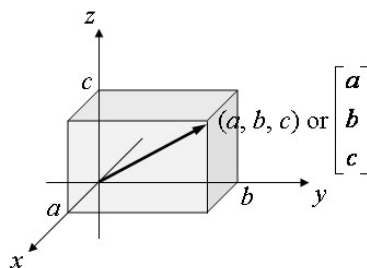


Figure 4.1.4. A vector in 3D rectangular coordinate system

Observe that any 2D vectors $\begin{bmatrix} a \\ b \end{bmatrix}$ can be regarded as a 3D vector $\begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$. Hence, throughout the sequel, the term 'vector' refers to 3D vector. We observe further that any vector $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ can be written as

$$\mathbf{v} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The three particular vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are called the **standard basis vectors** of the 3D rectangular coordinate system, denoted by $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$, respectively. These standard basis vectors have norm 1, and they are mutually perpendicular, as shown in Figure 4.1.5.

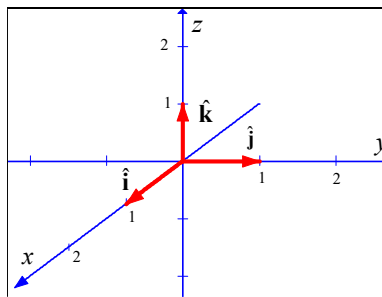


Figure 4.1.5. The standard basis vectors in 3D rectangular coordinate system

Thus, a vector $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ can be written as $\mathbf{v} = a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}}$; this is the **rectangular form** of $\mathbf{v}$.

Referring to Figure 4.1.4, the norm and direction of $\mathbf{v} = a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}}$ are given as follows.

$$(3D) \text{ Norm: } \|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2} \quad (4.4)$$

$$(3D) \text{ Direction: } \cos \theta_x = \frac{a}{\|\mathbf{v}\|}, \quad \cos \theta_y = \frac{b}{\|\mathbf{v}\|}, \quad \cos \theta_z = \frac{c}{\|\mathbf{v}\|} \quad (4.5)$$

where θ_x , θ_y , and θ_z are the angles between the line segment and x -, y -, and z - axes, respectively

The cosines in (4.5) are called directional cosines, each of which can be derived easily as shown in Figure 4.1.6.

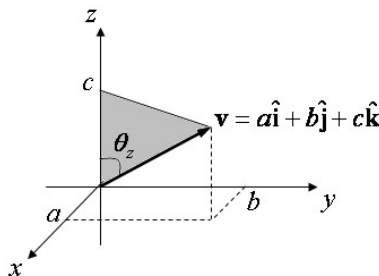


Figure 4.1.6. Direction angle of $\mathbf{v}$ with respect to z -axis

Example 4.1.1

The norm and unit vector of $\mathbf{v} = 5\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ are:

$$\|\mathbf{v}\| = \sqrt{5^2 + 2^2 + (-3)^2} = \sqrt{25 + 4 + 9} = \sqrt{38}$$

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{1}{\sqrt{38}} \begin{bmatrix} 5 \\ 2 \\ -3 \end{bmatrix}$$

When we are working solely on 2D vectors, we may omit the $\hat{\mathbf{k}}$ component and define $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ as the standard basis vectors of 2D rectangular coordinate system, as shown in Figure 4.1.7.

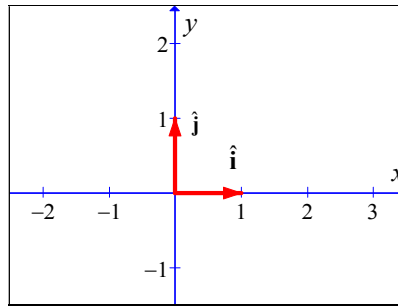


Figure 4.1.7. The standard basis vectors in 2D rectangular coordinate system

A directed line segment originated from the origin is called **position vector**; otherwise, it is called **direction vector**. Figure 4.1.8 shows the vector $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ as the position vector $\overrightarrow{OP}$ and as the direction vector $\overrightarrow{AB}$.

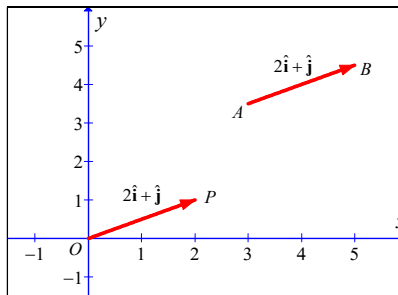


Figure 4.1.8 Position vector vs. direction vector

Note that for any points P and Q , we have $\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ} = \overrightarrow{OQ} - \overrightarrow{OP}$, as shown in Figure 4.1.9.

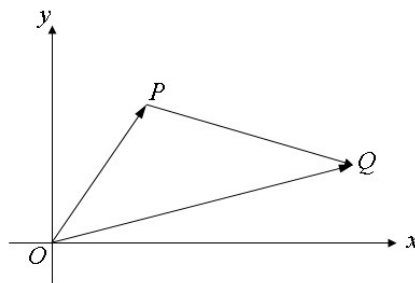


Figure 4.1.9. Direction vector $\overrightarrow{PQ}$ as $\overrightarrow{OQ} - \overrightarrow{OP}$

4.2 Dot Product and Cross Product

In the previous section, we have observed two algebraic operations on vectors: addition and scaling. We will discuss two other important operations: dot product and cross product, which are motivated from applications in classical mechanics.

Given two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, the dot product of $\mathbf{u}$ and $\mathbf{v}$ is the scalar P that represents the product of the norms of $\mathbf{u}$ and $\mathbf{v}$ along the direction of $\mathbf{u}$. This is done by observing the projection of $\mathbf{v}$ on $\mathbf{u}$, denoted by $\mathbf{v}'$, as shown in Figure 4.2.1.

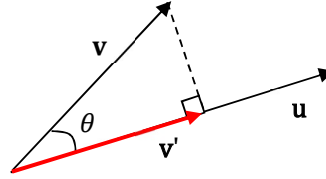


Figure 4.2.1. The projection of $\mathbf{v}$ on $\mathbf{u}$

If $\mathbf{v}'$ points to the same direction as $\mathbf{u}$, then $P = \|\mathbf{v}'\| \cdot \|\mathbf{u}\|$; otherwise, $P = -\|\mathbf{v}'\| \cdot \|\mathbf{u}\|$. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Observe that

$$\|\mathbf{v}'\| = \begin{cases} \|\mathbf{v}\| \cos \theta & \text{for } 0 \leq \theta < \frac{\pi}{2} \\ -\|\mathbf{v}\| \cos \theta & \text{for } \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

Hence, we obtain $P = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \cos \theta$. This quantity has meaningful geometric and physical interpretation only if both $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors.

Definition 4.2.1 (Dot Product)

Let θ be the angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$. The **dot product** $\mathbf{u} \cdot \mathbf{v}$ is a scalar as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \cos \theta$$

We use the formula below to compute the dot product of two nonzero vectors.

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 \quad (4.6)$$

The following algebraic laws apply for the dot product.

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (commutative)
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (distributive)

From Definition 4.2.1, we can use the dot product to calculate the angle θ between $\mathbf{u}$ and $\mathbf{v}$.

$$\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \cdot \|\mathbf{v}\|} \right) \quad (4.7)$$

Example 4.2.2

Given two vectors $\mathbf{u} = \begin{bmatrix} -3 \\ 4 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, the angle between $\mathbf{u}$ and $\mathbf{v}$ is

$$\begin{aligned}\theta &= \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \cdot \|\mathbf{v}\|} \right) \\ &= \cos^{-1} \left(\frac{(-3)(-1) + (4)(1) + (2)(1)}{\sqrt{(-3)^2 + 4^2 + 2^2} \cdot \sqrt{(-1)^2 + 1^2 + 1^2}} \right) = \cos^{-1} \left(\frac{9}{\sqrt{87}} \right) \approx 0.2657\end{aligned}$$

The following result is a direct consequence of the Formula (4.7).

Theorem 4.2.3

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

In 3D space, a vector that is perpendicular to a given plane is called **normal vector** of the plane. For example, the standard basis vector $\hat{\mathbf{k}}$ is a normal vector to the plane containing $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$, called the xy -plane. Obviously, the vectors $t\hat{\mathbf{k}}$ are also normal vectors to the xy -plane for any nonzero scalar t .

Given two nonparallel, nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, the cross product of $\mathbf{u}$ and $\mathbf{v}$ is a vector $\mathbf{w}$ that satisfies the following properties:

1. $\|\mathbf{w}\|$ is the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$, and
2. $\mathbf{w}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

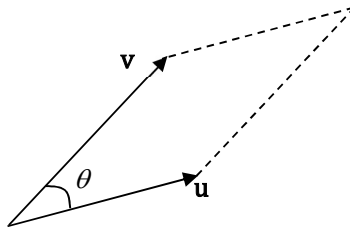


Figure 4.2.2. The parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$

Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. The area of the parallelogram in Figure 4.2.2 is $\|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \sin \theta$. So, $\|\mathbf{w}\| = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \sin \theta$.

A unit vector perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{v}$ is called a unit normal vector, denoted by $\hat{\mathbf{n}}$, as shown in Figure 4.2.3.

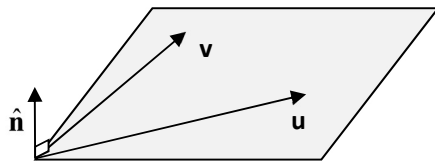


Figure 4.2.3. A unit normal vector $\hat{\mathbf{n}}$ perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{v}$

From this observation, the cross product of $\mathbf{u}$ and $\mathbf{v}$ is given as follows.

Definition 4.2.4 (Cross Product)

Let θ be the angle between two nonparallel, nonzero vectors $\mathbf{u}$ and $\mathbf{v}$. And let $\hat{\mathbf{n}}$ be a unit normal vector perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{v}$. The **cross product** $\mathbf{u} \times \mathbf{v}$ is a vector given as

$$\mathbf{u} \times \mathbf{v} = (\|\mathbf{u}\| \cdot \|\mathbf{v}\| \cdot \sin \theta) \hat{\mathbf{n}}$$

The direction of the unit normal vector $\hat{\mathbf{n}}$ in Definition 4.2.4 is determined by the right-hand rule as shown in Figure 4.2.4.

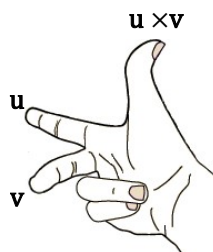


Figure 4.2.4. The right-hand rule to determine the direction of $\mathbf{u} \times \mathbf{v}$

To compute the cross product of two nonparallel, nonzero vectors $\mathbf{u} = u_1\hat{\mathbf{i}} + u_2\hat{\mathbf{j}} + u_3\hat{\mathbf{k}}$ and $\mathbf{v} = v_1\hat{\mathbf{i}} + v_2\hat{\mathbf{j}} + v_3\hat{\mathbf{k}}$, we use the following.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \quad (4.8)$$

The cross product of vectors is not commutative, but $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.

Example 4.2.5

Given $\mathbf{u} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{v} = -4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, a normal vector to both $\mathbf{u}$ and $\mathbf{v}$ is either $\mathbf{u} \times \mathbf{v}$ or $\mathbf{v} \times \mathbf{u}$.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \hat{\mathbf{i}} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \hat{\mathbf{k}} = -2\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 10\hat{\mathbf{k}}$$

Example 4.2.6

A unit vector perpendicular to the plane containing $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$ can be found by taking the cross product of any two vectors formed by these three points.

$$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \times \begin{bmatrix} -2 \\ 2 \\ 2 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}.$$

A unit normal vector $\hat{\mathbf{n}}$ to the plane is given as

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{1}{\sqrt{6^2 + 6^2}} \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

4.3 Lines in $\mathbb{R}^3$

We denote by $\mathbb{R}^2$ and $\mathbb{R}^3$ the collection of points in the 2D and 3D rectangular coordinate systems, respectively, i.e. $\mathbb{R}^2 = \{(x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$ and $\mathbb{R}^3 = \{(x, y, z) : x \in \mathbb{R}, y \in \mathbb{R}, \text{ and } z \in \mathbb{R}\}$. Any point in $\mathbb{R}^3$ can be represented by a triplet (a, b, c) .

Let f , g , and h be continuous functions of t . The equations

$$\begin{cases} x = f(t) \\ y = g(t) \\ z = h(t) \end{cases}$$

are called **parametric equations** with parameter t . The resulting curve is called parametric curve in $\mathbb{R}^3$. If the component $z = h(t)$ is omitted, then we have a parametric curve in $\mathbb{R}^2$ as shown in Example 4.3.1.

Example 4.3.1

- (a) $\begin{cases} x = t \\ y = 2t \end{cases}$ for $t \in \mathbb{R}$ is a straight line in $\mathbb{R}^2$; it is equivalent to $y = 2x$
- (b) $\begin{cases} x = \cos t \\ y = \sin t \end{cases}$ for $0 \leq t < 2\pi$ is a circle in $\mathbb{R}^2$; it is equivalent to $x^2 + y^2 = 1$
- (c) $\begin{cases} x = 2 \cos t \\ y = \sin t \end{cases}$ for $0 \leq t < 2\pi$ is an ellipse in $\mathbb{R}^2$; it is equivalent to $x^2 + 4y^2 = 4$

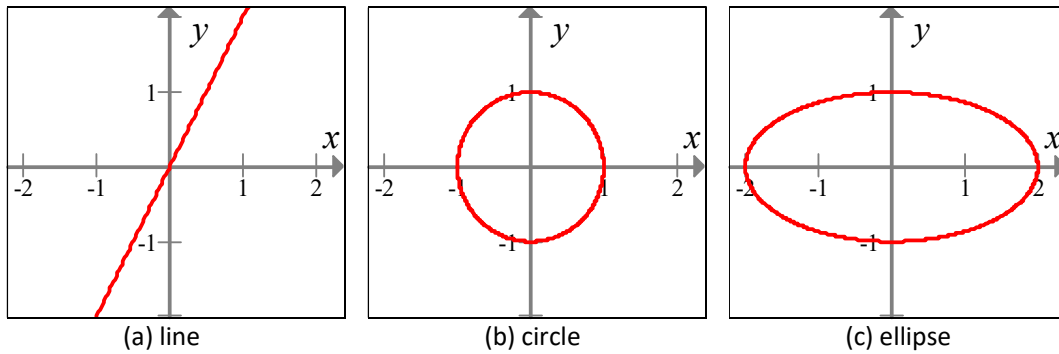


Figure 4.3.1. Parametric Curves in $\mathbb{R}^2$ from Example 4.3.1

We will find a parametric equations that represent a line L in $\mathbb{R}^3$ that is parallel to $\mathbf{v} = \begin{bmatrix} A \\ B \\ C \end{bmatrix}$ and passing through a point $Q(x_0, y_0, z_0)$. The given vector $\mathbf{v}$ acts like the gradient of a line in $\mathbb{R}^2$. Let $P(x, y, z)$ be any point on the line L as shown in Figure 4.3.2.

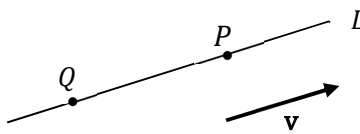


Figure 4.3.2. A line L that is parallel to the vector $\mathbf{v}$

Since L is parallel to $\mathbf{v}$, there is a scalar t such that the vector $\overrightarrow{QP}$ satisfies $\overrightarrow{QP} = t\mathbf{v}$, which implies

$$\begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = t \begin{bmatrix} A \\ B \\ C \end{bmatrix} \Rightarrow \begin{cases} x - x_0 = tA \\ y - y_0 = tB \\ z - z_0 = tC \end{cases}$$

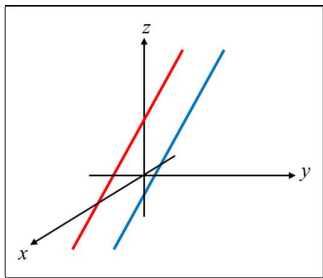
Note that each value of t corresponds to a point P on the line L . As t varies over all real numbers, the collection of all such points P will form the line L . The equations for a line in $\mathbb{R}^3$ that passes through $Q(x_0, y_0, z_0)$ and is parallel to the **direction vector** $\mathbf{v} = \begin{bmatrix} A \\ B \\ C \end{bmatrix}$ are given as follows.

Equations for a Line in $\mathbb{R}^3$

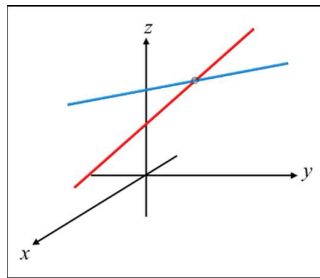
$$\text{Parametric: } \begin{cases} x = x_0 + At \\ y = y_0 + Bt \\ z = z_0 + Ct \end{cases} \text{ for } t \in \mathbb{R} \quad (4.9a)$$

$$\text{Symmetric: } \frac{x - x_0}{A} = \frac{y - y_0}{B} = \frac{z - z_0}{C} \quad (4.9b)$$

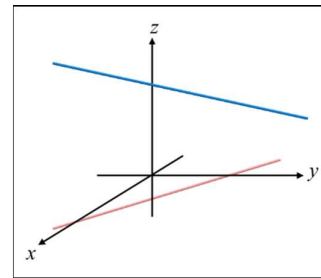
Two lines in $\mathbb{R}^3$ are parallel if their direction vectors are parallel; otherwise, they may intersect or skew, see Figure 4.3.3.



(a) parallel



(b) intersect



(c) skew

Figure 4.3.3. Two lines in $\mathbb{R}^3$

Example 4.3.3

Determine whether each of the following pair of lines are parallel or intersect or skew.

(a) $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{3}$ and $\frac{x+5}{4} = \frac{y-2}{-2} = \frac{z-3}{6}$

(b) $\begin{cases} x = 2t - 1 \\ y = 3 - t \\ z = 3t \end{cases}$ and $\begin{cases} x = 5 - t \\ y = 3t + 1 \\ z = 2t - 1 \end{cases}$

(a) The direction vectors of the first and second lines are $\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$.

Since $2\mathbf{v}_1 = \mathbf{v}_2$, the two lines are parallel.

(b) The direction vectors of the first and second lines are $\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$. The two lines are not parallel. Suppose they intersect at (a, b, c) . This gives a linear system of two unknowns:

$$\begin{cases} 2t - 1 = a = 5 - u \\ 3 - t = b = 3u + 1 \\ 3t = c = 2u - 1 \end{cases} \Rightarrow \begin{cases} 2t + u = 6 \\ t + 3u = 2 \\ 3t - 2u = -1 \end{cases},$$

which has no solution. Thus, the two lines are skew.

The distance from a point P to the line L is defined to be the shortest length of line segments $\overline{PQ}$ for all points Q on L ; it is given as follows.

Theorem 4.3.4 (Distance from a Point to a Line)

Let Q be any point on the line L with direction vector $\mathbf{v}$. Given a point $P(x_0, y_0, z_0)$, the distance d between P and L is given as

$$d = \|\overrightarrow{PQ} \times \hat{\mathbf{v}}\|$$

Proof.

Let θ be the angle between the vector $\overrightarrow{PQ}$ and the line L as shown in Figure 4.3.4. Observe that $d = \|\overrightarrow{PQ}\| \cdot \sin \theta$. Now, consider the parallelogram spanned by the vectors $\overrightarrow{PQ}$ and $\mathbf{v}$. The area of the parallelogram is given as follows.

$$\|\overrightarrow{PQ} \times \mathbf{v}\| = \|\overrightarrow{PQ}\| \cdot \|\mathbf{v}\| \cdot \sin \theta = \|\mathbf{v}\| \cdot d \Rightarrow d = \frac{\|\overrightarrow{PQ} \times \mathbf{v}\|}{\|\mathbf{v}\|} = \|\overrightarrow{PQ} \times \hat{\mathbf{v}}\|$$

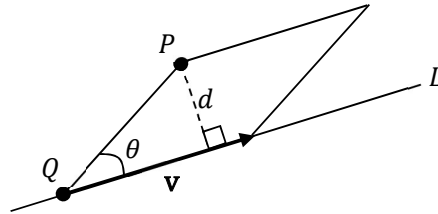


Figure 4.3.4. Distance between the point P and the line L

Example 4.3.5

Find the distance between the point $P(1, 2, -3)$ and the line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{3}$.

The point $Q(1, -3, 0)$ is on the line. Then $\overrightarrow{PQ} \times \mathbf{v} = \begin{bmatrix} 0 \\ -5 \\ 3 \end{bmatrix} \times \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & -5 & 3 \\ 2 & -1 & 3 \end{vmatrix} = \begin{bmatrix} -12 \\ 6 \\ 10 \end{bmatrix}$.

Thus,

$$d = \frac{\|\overrightarrow{PQ} \times \mathbf{v}\|}{\|\mathbf{v}\|} = \frac{\sqrt{(-12)^2 + 6^2 + 10^2}}{\sqrt{2^2 + (-1)^2 + 3^2}} = \frac{\sqrt{280}}{\sqrt{14}} = \sqrt{20} = 2\sqrt{5} \text{ units.}$$

4.4 Planes in $\mathbb{R}^3$

In $\mathbb{R}^3$, the equation $z = 0$ describes the collection of points

$$\{(x, y, z) \in \mathbb{R}^3 : z = 0\} = \{(x, y, 0) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\},$$

which is a plane, called the xy -plane, as shown in Figure 4.4.1. Similarly, the equation $x = 0$ describes yz -plane, and the equation $y = 0$ describes xz -plane.

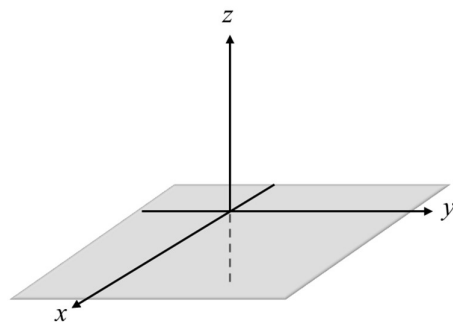


Figure 4.4.1. xy -plane represented by $z = 0$

A vector $\mathbf{n}$ is called **normal vector** to a given plane if it is orthogonal (i.e. perpendicular) to every vector on the plane, i.e. $\mathbf{n} \cdot \mathbf{v} = 0$ for all vectors $\mathbf{v}$ on the plane.

Suppose we have a plane in $\mathbb{R}^3$ containing the point $Q(x_0, y_0, z_0)$ and has a normal vector $\mathbf{n} = \begin{bmatrix} A \\ B \\ C \end{bmatrix}$.

Then for any point $P(x, y, z)$ on the plane, $\mathbf{n} \cdot \overrightarrow{QP} = 0 \Rightarrow \begin{bmatrix} A \\ B \\ C \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$, which implies

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \quad (4.10)$$

Equation (4.10) is called the **point-normal form** of the plane equation and the numbers $[A, B, C]$ are called the **attitude numbers** of the plane. The point-normal form of the plane equation can be rearranged to yield $Ax + By + Cz + D = 0$, which is the **standard form** of the plane equation.

Two planes are parallel if their attitude numbers are proportional. Otherwise, the intersection of these planes is a line in $\mathbb{R}^3$.

Note that normal vector of a plane is not unique. If $\mathbf{n}$ is a normal vector to a given plane, then any vector parallel to $\mathbf{n}$ is also a normal vector to the plane.

Given three points P, Q , and R on a plane in $\mathbb{R}^3$, we can find a normal vector to the plane by taking cross product $\overrightarrow{PQ} \times \overrightarrow{PR}$ or $\overrightarrow{QP} \times \overrightarrow{QR}$ or $\overrightarrow{RP} \times \overrightarrow{RQ}$.

Example 4.4.1

Find the standard equation of the plane containing $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

From Example 4.2.6, a normal vector $\mathbf{n}$ to the plane is given as $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}$. So, the standard equation of the plane is $6(x - 1) + 6(z - 0) = 0 \Rightarrow x + z - 1 = 0$.

Theorem 4.4.2 (Distance from a Point to a Plane)

Let $\mathbf{n}$ be a normal vector to the plane $Ax + By + Cz + D = 0$ containing a point Q . Given a point $P(x_0, y_0, z_0)$, the distance d between P and the plane is given as

$$d = |\overrightarrow{PQ} \cdot \hat{\mathbf{n}}| = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

Proof.

Let θ be the angle between the vector $\overrightarrow{QP}$ and the normal vector $\mathbf{n}$ as shown in Figure 4.4.2. Observe that $d = \|\overrightarrow{QP}\| \cdot \cos \theta$. Now, consider the absolute value of the dot product of $\overrightarrow{QP}$ and $\mathbf{n}$.

$$|\overrightarrow{PQ} \cdot \mathbf{n}| = \|\overrightarrow{QP}\| \cdot \|\mathbf{n}\| \cdot |\cos \theta| = \|\mathbf{n}\| \cdot d \Rightarrow d = \frac{|\overrightarrow{PQ} \cdot \mathbf{n}|}{\|\mathbf{n}\|} = |\overrightarrow{PQ} \cdot \hat{\mathbf{n}}|$$

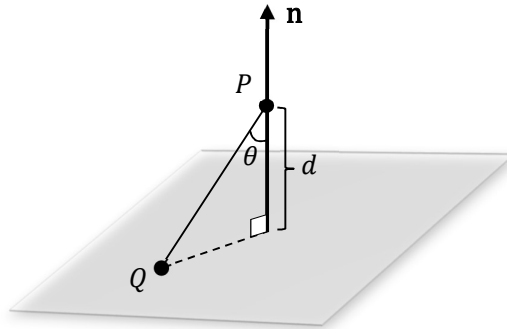


Figure 4.4.2. Distance from a point to a plane

Moreover, since $Q(x_1, y_1, z_1)$ satisfies $Ax_1 + By_1 + Cz_1 + D = 0$, we have

$$\begin{aligned} \overrightarrow{PQ} \cdot \mathbf{n} &= \begin{bmatrix} x_0 - x_1 \\ y_0 - y_1 \\ z_0 - z_1 \end{bmatrix} \cdot \begin{bmatrix} A \\ B \\ C \end{bmatrix} = A(x_0 - x_1) + B(y_0 - y_1) + C(z_0 - z_1) \\ &= Ax_0 + By_0 + Cz_0 - (Ax_1 + By_1 + Cz_1) \\ &= Ax_0 + By_0 + Cz_0 + D. \end{aligned}$$

Example 4.4.3

The distance between the point $P(1, 2, -3)$ and the plane $2x - 2y - z + 5 = 0$ is

$$d = \frac{|2(1) - 2(2) - (-3) + 5|}{\sqrt{2^2 + (-2)^2 + (-1)^2}} = \frac{6}{\sqrt{9}} = 2 \text{ units.}$$

4.5 Quadric Surfaces

A **quadric surface** in $\mathbb{R}^3$ is the graph of an equation of the form

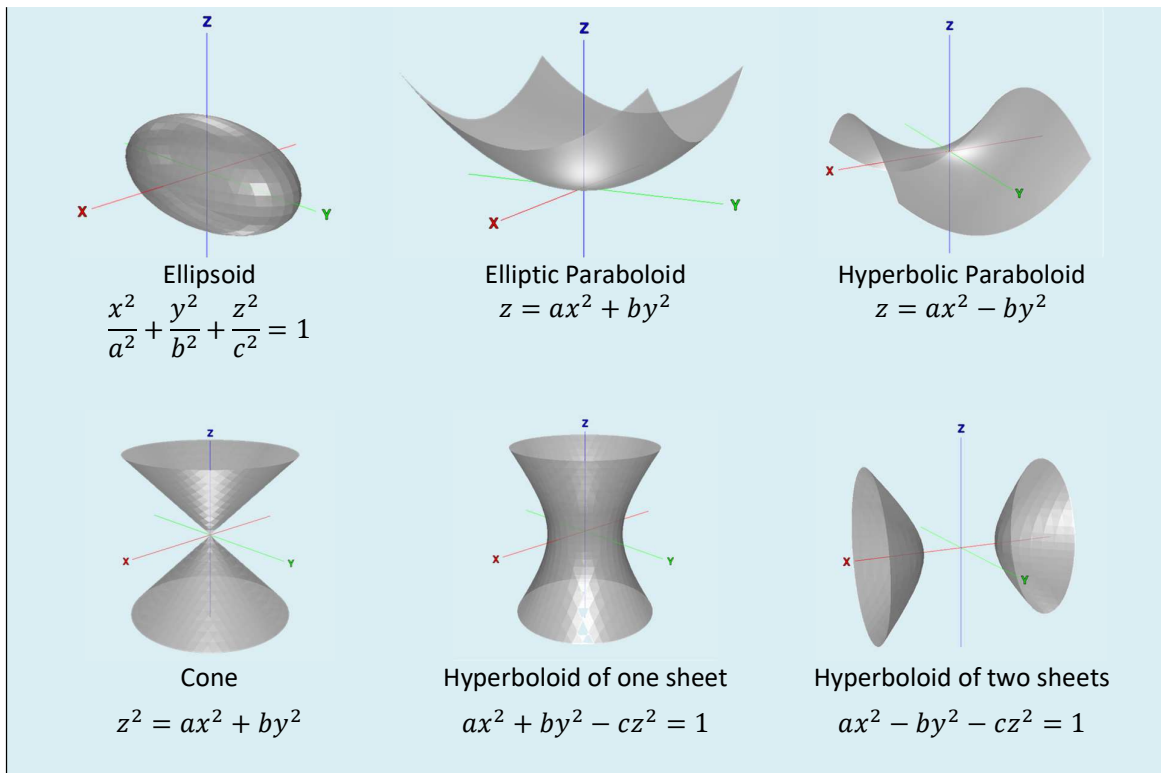
$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$

where $A, B, C, D, E, F, G, H, I$, and J are constants. This equation is called a second-degree equation in three variables x, y , and z . The standard forms of the equations of quadric surfaces are given as follows.

Standard Form 1:	$z = Mx^2 + Ny^2$	(4.11)
Standard Form 2:	$Px^2 + Qy^2 + Rz^2 = S$	(4.12)

Quadric surfaces having the first standard form are either elliptic paraboloid or hyperbolic paraboloid. Common quadric surfaces are shown below (a, b, c are all positive).

Quadric Surfaces



The simplest quadric surface is a sphere, a special case of ellipsoid with $a = b = c$. A **sphere** with center $P(a, b, c)$ and radius $r > 0$ is the collection of all points in $\mathbb{R}^3$ with distance r from P . In $\mathbb{R}^3$, the distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, denoted by $|AB|$, is

$$|AB| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2},$$

which is equivalent to $\|\overline{AB}\|$, norm of a 3D vector.

If $Q(x, y, z)$ is a point on the sphere, then $|PQ| = r \Rightarrow \sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} = r$. Simplifying, we obtain the equation of the sphere with center (a, b, c) and radius r :

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2.$$

Example 4.5.1

A quadric surface of the form $x^2 + y^2 + z^2 + 2x - 2y + 4z - 3 = 0$ can be written as

$$(x^2 + 2x + 1) + (y^2 - 2y + 1) + (z^2 + 4z + 4) = 9 \Rightarrow (x+1)^2 + (y-1)^2 + (z+2)^2 = 3^2,$$

which is a sphere with center $(-1, 1, -2)$ and radius 3. By translating the center of the sphere into the origin, we obtain $x^2 + y^2 + z^2 = 3^2$, the second standard form.

4.6 Exercises

1. Calculate the following dot and cross products of $\mathbf{u}$ and $\mathbf{v}$.

(a) $\mathbf{u} = \begin{bmatrix} 2 \\ -4 \\ \sqrt{5} \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ 4 \\ \sqrt{5} \end{bmatrix}$

(d) $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(b) $\mathbf{u} = \begin{bmatrix} 1.5 \\ -0.5 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

(e) $\mathbf{u} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{k}}$ and $\mathbf{v} = \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

(c) $\mathbf{u} = \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

(f) $\mathbf{u} = 2\hat{\mathbf{i}}$ and $\mathbf{v} = -3\hat{\mathbf{j}}$

2. Find the parametric and symmetric equations of each of the following lines.

(a) The line through the point $P(3, -4, -1)$ parallel to the vector $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$.

(b) The line through $P(1, 2, -1)$ and $Q(-1, 0, 1)$.

(c) The line through $(2, 3, 4)$ and perpendicular to the plane $3x + 7y - 2z = 6$.

3. Determine whether each the following pair of lines are parallel, intersect, or skew.

(a) $\begin{cases} x = 1 + 2t \\ y = 3 - t \\ z = 3t \end{cases}$ and $\begin{cases} x = 3 + 4t \\ y = 5 - 2t \\ z = 6t \end{cases}$

(d) $\begin{cases} x = 3 + 2t \\ y = -1 + 4t \\ z = 2 - t \end{cases}$ and $\begin{cases} x = 3 + 2t \\ y = 2 + t \\ z = -2 + 2t \end{cases}$

(b) $\begin{cases} x = t - 1 \\ y = 3 + t \\ z = t \end{cases}$ and $\begin{cases} x = 1 - t \\ y = 3 - t \\ z = -t \end{cases}$

(e) $\begin{cases} x = 1 + 2t \\ y = -1 - t \\ z = 3t \end{cases}$ and $\begin{cases} x = 2 - t \\ y = 3t \\ z = 1 + t \end{cases}$

(c) $\begin{cases} x = 3 + 2t \\ y = -1 + 4t \\ z = 2 - t \end{cases}$ and $\begin{cases} x = 1 + 4t \\ y = 1 + 2t \\ z = -3 + 4t \end{cases}$

(f) $\begin{cases} x = 2 - t \\ y = 3t \\ z = 1 + t \end{cases}$ and $\begin{cases} x = 5 + 2t \\ y = 1 - t \\ z = 8 + 3t \end{cases}$

4. Find the standard equation of each of the following planes.
- (a) The plane through $P(0, -2, -1)$ normal to $\mathbf{n} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$
 - (b) The plane through $(1, -1, 3)$ parallel to the plane $3x + y + z = 7$
 - (c) The plane through $(1, 1, -1)$, $(2, 0, 2)$, and $(0, -2, 1)$
 - (d) The plane through $(2, 4, 5)$, $(1, 5, 7)$, and $(-1, 6, 8)$
 - (e) The plane through $P(2, 4, 5)$ perpendicular to the line $\begin{cases} x = 5 + t \\ y = 1 + 3t \\ z = 4t \end{cases}$
 - (f) The plane through $A(1, -2, 1)$ perpendicular to the vector from the origin to A .
5. Find the distance of each of the following.
- (a) The point $(0, 0, 12)$ and the line $\begin{cases} x = 4t \\ y = -2t \\ z = 2t \end{cases}$
 - (b) The point $(2, 1, 3)$ and the line $\begin{cases} x = 2 + 2t \\ y = 1 + 6t \\ z = 3 \end{cases}$
 - (c) The point $(0, 1, 3)$ and the plane $2x + y + 2z = 4$
 - (d) The line $\begin{cases} x = 2 + t \\ y = 1 + t \\ z = -0.5 - 0.5t \end{cases}$ and the plane $x + 2y + 6z = 10$
6. Determine the name of each of the following quadric surfaces.
- (a) $9x^2 - y^2 - z^2 = 9$
 - (b) $9x^2 + y^2 + z^2 = 9$
 - (c) $z = x^2 + 4y^2$
 - (d) $x^2 + 4y^2 - z^2 = 1$
 - (e) $z = x^2 - 4y^2$
 - (f) $z^2 = x^2 + 4y^2$
7. Given two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, let $\mathbf{v}'$ be the projection vector of $\mathbf{v}$ on $\mathbf{u}$. Show that

$$\mathbf{v}' = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2} \right) \mathbf{u}$$

